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| **Fiscal Year:** | FY 2015 | **Task Last Updated:** | FY 06/02/2015 |
| **PI Name:** | Alperin, Noam Ph.D. | | |
| **Project Title:** | Retrospective Analyses of MR Imaging of Cerebrospinal Fluid (CSF) Dynamics Pre and Post Space Flight | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Biomedical countermeasures | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HHC**:Human Health Countermeasures | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 33146-1269 | **Congressional District:** | 27 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | Directed Research |
| **Start Date:** | 12/01/2013 | **End Date:** | 06/30/2017 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Norsk, Peter | **Contact Phone:** | |
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| **Flight Program:** | Shuttle/ISS | | |
| **Flight Assignment:** | Pre- and Post-Flight<br>NOTE: Extended to 6/30/2017 per A. Allcorn/JSC and NSSC information (Ed., 3/16/17)<br>NOTE: Extended to 2/28/2017 per NSSC information (Ed., 8/25/16)<br>NOTE: Extended to 8/31/2016 per NSSC information (Ed., 9/9/15)<br>NOTE: End date is now 8/31/2015, per PI and NSSC information (Ed., 3/27/15)<br>NOTE: End date is now 9/30/2015 (originally 11/30/2014) per J. Villarreal/HRP (Ed., 9/30/14)<br>NOTE: End date is now 6/30/2015 (originally 11/30/2014) per R. Brady/HRP (Ed., 9/15/14)<br>NOTE: End date is now 3/30/2015 (originally 11/30/2014) per R. Brady/HRP (Ed., 5/2/14) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Barr, Yael M.D. ( NASA Johnson Space Center Bioastronautics / UTMB ) | | |
| **Grant/Contract No.:** | NNX14AB51G | | |
| **Performance Goal No.:** | | | |

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| **Performance Goal Text:** | |
| **Task Description:** | Microgravity-induced elevated intracranial pressure (ICP) is currently thought to be the cause of the Visual Impairment Intracranial Pressure (VIIP) syndrome often reported in astronauts during and after long duration space flights. This hypothesis is supported by several postflight measurements of ICP obtained via a lumbar puncture (LP). The cause for ICP elevation in-flight has not yet been established. Current investigations of the VIIP syndrome focus on the vascular and interstitial cephalad fluid redistribution as a primary cause for elevated ICP. The potential contribution of cephalad cerebrospinal fluid (CSF) shift has not yet been investigated and is the focus of this study. As the cranium contains more CSF than blood, this component of fluid shift may be a larger contributor to ICP elevations than the vascular/interstitial component. The potential reduction in spinal canal compliance buffering due to spinal elongation might contribute to ICP elevations as well.<br>The proposed study will utilize specialized MRI sequences, collected in short-duration and long-duration crewmembers, to assess CSF hydrodynamics and vascular hemodynamics. Key physiological parameters that will be calculated from the MRI data include total cerebral blood flow, venous drainage, intracranial compliance and pressure as well as cranio-spinal compliance distribution. These parameters will be tested for association with changes in visual/ocular function and morphology, and measures of spinal elongation. Effects of mission duration will be elucidated by comparing short- and long-duration crewmembers. The variable time period between landing and the MRI studies is a potential confounding factor of this study. |
| **Rationale for HRP Directed Research:** | This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | This is a retrospective study that focuses on quantitative analyses of specialized MRI (magnetic resonance imaging) scans performed on NASA crewmembers before and shortly after completion of their short and long-duration spaceflights. The specialized MRI scans were obtained between October 2010 and early 2014 and include 25 subjects, 10 who flew short term missions on the space shuttle, and 15 who flew long-duration (5-6 months) missions on the International Space Station (ISS).<br>The specialized MRI scan protocol included dynamic imaging of the pulsatile blood and CSF flow to and from the cranium during the cardiac cycle from which important cerebral hemodynamics and cranio-spinal hydrodynamics measures can be calculated [1-2]. The study aims to compare pre-to-post flight changes between the short and long duration spaceflight cohorts. Second, the study attempt to identify association between the magnitude pre-to-post flight changes in specific quantitative MRI measures and the presence and severity of signs and symptoms indicative of VIIP syndrome.<br><br>The main cerebral hemodynamics and quantitative cranio-spinal hydrodynamics measures include 1) total cerebral blood flow, 2) venous outflow, 3) cranio-spinal CSF flow and stroke volume, 4) cranio-spinal compliance distribution, and 5) MR-derived ICP (MRICP). The VIIP Clinical Practice Guideline (CPG) classification is being used initially as a measure for VIIP syndrome severity.<br><br>Efforts during the first award period were focused toward 1) compliance with IRB (Institutional Review Board) requirements, 2) defining, and 3) collecting crewmembers' clinical background information obtained before during and following spaceflight. To date, consent has been obtained from 17 of the 25 subjects participating in the study. Therefore, in order to facilitate the research progress, initial imaging and clinical data analyses will be done using the sub-group of crewmembers for whom a concern has been obtained. Furthermore, a no cost extension has been requested and granted till 8/30/2015 to allow for the additional time needed for the data gathering under the IRB guidelines.<br><br>We have been working closely with the Lifetime Surveillance of Astronaut Health (LSAH) to facilitate and streamline the process as much as possible. Data for all 25 subjects will be compiled at the same time; LSAH will release the first 17 consented subjects' data in 5 phases. Concurrently, LSAH will pursue informed consent of the remaining 8 subjects. Currently the data release is schooled to be delivered in 6 waves. Current estimate for completion of clinical data delivery from LSAH is end of November 2014.<br><br>*Consent Timeline notes: For the 5 International Partner (IP) subjects, consenting cannot begin until the research protocol is approved by each subject's IP agency and Human Research Multilateral Review Board (HRMRB), and agreements have been reached among the agencies for implementation of data sharing.<br><br>Preliminary Analyses<br><br>Preliminary results are based on analyses of data from 17 crewmembers that have been recently consented. The initial analysis compares post-to-pre changes in total cerebral blood flow (CBF) and cranio-spinal CSF flow volume. Total CBF is obtained by summation of the mean flow rates through the 4 blood vessels supplying the brain (right and left internal carotid and vertebral arteries). Volumetric flow rates were obtained using an automated lumen segmentation technique shown to have 3-4-fold improved reproducibility and accuracy over manual lumen segmentation [3]. Two cohorts, 5 short-duration and 8 long-duration crewmembers, who were scanned within 3 to 8 days post landing were included.<br><br>REFERENCES<br><br>[1] Alperin N. et al (2000) Radiology 217, 877–885. [2] Tain RW. et al (2011) J Magn Reson Imag 34,1397-1404. [3] Alperin N. et al (2005) Magn Reson Med 49,934–944. [4] Alperin N. et al (2013) AJNR Am J Neuroradiol 34, 2354-2359.<br><br>[Ed. note: compiled from PI's FY2015 annual report] |
| **Bibliography Type:** | Description: (Last Updated: ) |